9. Travelling Salesman Problem (TSP)

Aim:

To write a Python program to solve the Travelling Salesman Problem using a search algorithm.

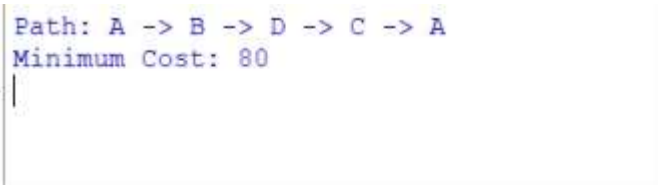
Algorithm:

1. Start the program.
2. Define the cities and their distances.
3. Select the starting city.
4. Generate possible routes to visit all cities.
5. Calculate the total distance of each route.
6. Return to the starting city.
7. Find the route with minimum total distance.
8. Display the shortest route and distance.
9. Stop.

INPUT

```
from itertools import permutations

cities = ['A', 'B', 'C', 'D']

cost = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

best = float('inf')
path = None

for p in permutations(range(1, 4)):
    route = (0,) + p + (0,)
    total = sum(
        cost[route[i]][route[i+1]]
        for i in range(len(route)-1)
    )

    if total < best:
        best = total
        path = route

print("Path:", " -> ".join(cities[i] for i in path))
print("Minimum Cost:", best)
```

OUTPUT



Result:

Thus, the Python program to solve the Travelling Salesman Problem was executed successfully.